

David Yi

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EDUCATION

Carnegie Mellon University

Expected May 2029

B.S. Statistics & Machine Learning; Minors in Robotics and Biomedical Engineering

Relevant coursework: Introduction to Human-Robot Interaction (16-467), Principles of Imperative Computation (15-122), Fundamentals of Programming & Computer Science (15-112), Matrices & Linear Transformations (21-241), Probability & Statistical Inference (36-235), Methods for Statistics & Data Science (36-202), Calculus in 3-D (21-259), Concepts of Mathematics (21-127); Introduction to Machine Learning (10-301) and General Robotics (16-281) in Spring 2027

TECHNICAL SKILLS

Languages: Python, C

Robotics & ML: ROS2, PyTorch, YOLO, OpenCV, Intel RealSense, Kinova Kortex API, NVIDIA Jetson, MoveIt

Tools: Linux, Git, Roboflow, SAM2, LaTeX

RESEARCH EXPERIENCE

Carnegie Mellon University Robotics Institute

Jan 2026 – Present

Undergraduate Researcher — RCHI and HARP labs; advised by Zackory Erickson and Henny Admoni

- Developing autonomous manipulation pipelines for a wheelchair-mounted Kinova Gen3 arm, working with RAMMP (Robotics and Mobility for Marginalized Populations) and ATDev.
- Building a robot-assisted feeding demonstration with a team of undergraduates, specifically responsible for the microwave open-and-close task, targeting December 2026.
- Stabilized the ROS2 environment for the feeding stack across Jetson and desktop machines, resolving dependency, version, and driver conflicts.
- Beginning work on eye-gaze intent inference for shared-autonomy manipulation.

Rutgers University

2023 – 2025

Independent Researcher — Topological Quantum Matter Engineering Lab

- Characterized the surfaces of topological insulators using an NT-MDT atomic force microscope, comparing bismuth chalcogenides Bi₂Te₃ and Bi₂Se₃ to explain why the former exhibits single-domain properties.
- Co-authored a preprint on twin-boundary formation in bismuth chalcogenides (arXiv:2410.21735).

PROJECTS

Autonomous Water-Bottle Pick and Handoff *Kinova Gen3, ROS2, Jetson Orin Nano*

2026

- Built an end-to-end pipeline taking a wheelchair-mounted arm from camera frame to grasp: detection, 3D localization, coordinate transforms, reachability checking, and handoff to a person.
- Fine-tuned yolo11s-seg to ~0.93 mIoU at 12–18 FPS on a Jetson Orin Nano after benchmarking YOLO11 and YOLO26 segmentation variants; built and labeled the ~350-image training dataset using Roboflow and SAM2.
- Performed hand-eye calibration and depth-to-3D projection; integrated an Intel RealSense camera as a fallback after the arm's onboard vision module failed.
- Migrated arm control from MoveIt to the Kortex API for real-time performance, adding EMA pose filtering, track-locking, retry logic, and pause/resume services for safe handoff.

LEADERSHIP & ACTIVITIES

Neurodiversity Ambassadors *Founder & President*

2022 – 2025

- Founded and led a student organization advocating for neurodivergent students in the Bridgewater, NJ school district.

CMU Club Hockey *Scheduling Officer*

2025 – Present

Gong Cha *Shift Lead*

2024 – Present

- Train new staff, manage closing operations and cash handling; maintained 10–13 hours per week alongside full-time coursework.

TookTook98 *Server & Kitchen Staff*

Summer 2026